

Zak M. Kassas

Contact Information	2015 Neil Ave., Office 420 Electrical & Computer Engineering ElectroScience Laboratory (ESL) The Ohio State University Columbus, OH 43210, USA	Office: (614) 247-6306 E-mail: zkassas@ieee.org Web: http://ece.osu.edu/people/kassas.2 Lab: http://ece.osu.edu/aspin Center: http://utc.engineering.osu.edu
Research Interests	Navigation systems, low Earth orbit satellites, cognitive sensing and software-defined radio, intelligent transportation systems, autonomous vehicles, cyber-physical systems	
Capsule Bio	Prof. Kassas is the TRC Endowed Chair in Intelligent Transportation Systems and a Full Professor of Electrical & Computer Engineering at The Ohio State University (OSU). He is also the Director of U.S. Department of Transportation Center: CARMEN (<u>C</u> enter for <u>A</u> utomated Vehicles <u>R</u> esearch with <u>M</u> ultimodal Assur <u>E</u> d <u>N</u> avigation), focusing on navigation resiliency and security of highly automated transportation systems.	

He received from President Biden the Presidential Early Career Award for Scientists and Engineers (PECASE), the highest honor bestowed by the U.S. government on outstanding scientists and engineers early in their careers. He is a Fellow of the IEEE and the Institute of Navigation (ION) and a Distinguished Lecturer of IEEE Aerospace & Electronic Systems Society and IEEE Intelligent Transportation Systems Society. He authored 200+ peer-reviewed journal and conference papers, 12 magazine articles, 3 invited book chapters, and 24 U.S. patents. He was ranked by ScholarGPS as the top scholar in the world in the field of Navigation. His awards include IEEE Harry Rowe Mimno Award, IEEE Signal Processing Society grand prize for beamforming research, ION Col. Thurlow Award, ION Burka Award, IEEE Walter Fried Award, Air Force Office of Scientific Research (AFOSR) Young Investigator Program (YIP) award, Office of Naval Research (ONR) YIP award, and National Science Foundation (NSF) CAREER award. His students have won several awards, including three Best Ph.D. Dissertation awards (from IEEE, ION, OSU); two US DOT Graduate Student of the Year awards; and 35+ best paper, student paper, and paper presentation awards.

He started his academic career in 2014 at University of California, Riverside; then was an Associate Professor at University of California, Irvine; then was very early promoted to Full Professor and joined The Ohio State University in 2022. His research has attracted \$27+ million in competitive grants from ONR, NSF, AFOSR, NASA, DOT, NIST, Sandia National Labs, Aerospace Corporation, among others. He has given 120+ invited presentations, keynotes, and plenaries, and served as a subject matter expert to DOD, GAO, DOT, and NSF. His research was featured in dozens of international media outlets (Science, BBC, Forbes, IEEE, ACM, among others) and appeared on 7 magazine covers. He is a Senior Editor of IEEE T-ITS and Associate Editor of IEEE T-AES. He organized and chaired numerous conferences, workshops, and special sessions: IEEE/ION PLANS, IEEE VTC, IEEE ICASSP, IEEE ITSC, ION GNSS+, ION ITM, FUSION, and ACC.

Prior to his academic career, he was a research & development engineer with the Control Design & Dynamical Systems Simulation Group at National Instruments (NI), where he contributed to the full-development cycle (research, implementation, and testing) of 6 major and minor releases of: LabVIEW Control Design & Simulation Module, LabVIEW System Identification Toolkit, and LabVIEW PID & Fuzzy Logic Toolkit.

Education	The University of Texas at Austin , Austin, TX	
	Ph.D. , Electrical & Computer Engineering, May 2014	
	Dissertation: Analysis and Synthesis of Collaborative Opportunistic Navigation Systems	
	Advisors: Todd E. Humphreys (primary) and Ari Arapostathis (co-advisor)	
	The University of Texas at Austin , Austin, TX	
	M.S.E. , Aerospace Engineering, August 2010	
	Report: Optimal $\mathcal{H}_2$ and $\mathcal{H}_\infty$ Control of Large Segmented Telescopes	
	Advisor: Robert H. Bishop	
	The Ohio State University , Columbus, OH	
	M.S. , Electrical & Computer Engineering, December 2003	
	Thesis: An Optimal Nonlinear Bayesian Filter Design and Combined Hospitability and Synthetic Inclination Approach for Tracking	
	Advisor: Ümit Özgüner	
	Lebanese American University , Byblos, Lebanon	
	B.E. with Honors, Electrical Engineering, March 2001	
	Final Project: A Power Matching Approach for Using GPS as a Reliable Standalone Positioning System	
	Advisor: Samer S. Saab	
Current Employment	TRC Endowed Chair in Intelligent Transportation Systems	Jul. 2023 – present
	The Ohio State University	
	Full Professor	Jul. 2022 – present
	The Ohio State University Department of Electrical & Computer Engineering	
Past Employment	Center Director	Sep. 2020 – present
	US Department of Transportation (USDOT) Tier 1 University Transportation Center (UTC) – CARMEN: <u>C</u> enter for <u>A</u> utomated Vehicles <u>R</u> esearch with <u>M</u> ultimodal Assur <u>E</u> d <u>N</u> avigation	
	Assistant/Associate Professor (Tenured)	Jan. 2019 – Jun. 2022
	University of California, Irvine Department of Mechanical & Aerospace Engineering Department of Electrical Engineering & Computer Science Institute of Transportation Studies	
	Assistant Professor	Jul. 2014 – Dec. 2018
	University of California, Riverside Department of Electrical & Computer Engineering	
	Graduate Research Assistant	Jun. 2011 – Jun. 2014
	The University of Texas at Austin Radionavigation Laboratory Wireless Networking & Communications Group (WNCG)	
	Adjunct Professor	Jan. – May, 2008 – 2011
	Texas State University, San Marcos, TX Ingram School of Engineering	

Research and Development (R&D) Engineer	Oct. 2004 – Dec. 2010
National Instruments Corp., Austin, TX	
Control Design & Dynamical Systems Simulations Group	
Graduate Research Associate	Apr. 2002 – Sep. 2004
The Ohio State University	
Collaborative Center of Control Science (CCCS)	
Laboratory Research Assistant	Mar. 2001 – May 2001
Lebanese American University, Byblos, Lebanon	
GPS Laboratory	

Research Grants

Total: \$27,775,150; Conducted as PI: \$24,585,450; Kassas' Share: \$10,157,075

Federal Grants

- **Title:** Development and Demonstration of GPS Alternative for Enhanced Security of the National Airspace
Sponsor: NASA
Amount: \$963,000
Period: 2024–2026
PI: Z. Kassas
- **Title:** Icarus: High Altitude Opportunistic Radio Frequency Navigation
Sponsor: Sandia National Laboratories
Amount: \$150,000
Period: 2023–2024
PI: Z. Kassas
- **Title:** Center for Automated Vehicles Research with Multimodal AssurEd Navigation (CARMEN+) – Tier 1 University Transportation Center (UTC)
Sponsor: US Department of Transportation (USDOT)
Amount: \$10,000,000 (Cost Share: \$5,000,000); **Kassas' Share: \$1,463,345** (Cost Share: \$767,610)
Period: 2023–2029
PI: Z. Kassas; **Co-PIs:** Q. Ahmed (OSU), C. Chen (OSU), D. Grejner-Brzezinska (OSU), Z. Lin (OSU), Ü. Özgüner (OSU), F. Özgüner (OSU), K. Redmill (OSU), C. Toth (OSU), A. Yener (OSU), C. Bhat (UT-Austin), T. Humphreys (UT-Austin), A. Chen (UCI), V. Deguzman (UCI), R. Jayakrishnan (UCI), S. Ritchie (UCI), A. Homaifar (NCAT), M. Mahmoud (NCAT)
- **Title:** Alternative PNT using Signals of Opportunity Navigation
Sponsor: Aerospace Corporation
Amount: \$350,000
Period: 2023–2025
PI: Z. Kassas
- **Title:** Cognitive Opportunistic Navigation with Unknown LEO and MEO Satellite Signals
Sponsor: Office of Naval Research (ONR)
Amount: \$750,000
Period: 2022–2026
PI: Z. Kassas

- **Title:** SMAC-FIRE: Closed-Loop Sensing, Modeling and Communications for Wild-FIRE
Sponsor: National Science Foundation (NSF)
Amount: \$1,199,087, **Kassas' Share:** **\$269,600**
Period: 2022–2025
PI: A. Swindlehurst (UCI), **Co-PI:** **Z. Kassas**, H. Jafarkhani (UCI), T. Banerjee (UCI), and J. Coen (National Center for Atmospheric Research)
- **Title:** Young Investigator Program (YIP): Dynamic Data Driven C-SPAN: Cognitive Sensing, Perception, Autonomy, and Navigation
Sponsor: Air Force Office of Scientific Research (AFOSR)
Amount: **\$450,000**
Period: 2022–2025
PI: **Z. Kassas**
- **Title:** Harnessing Terrestrial and Space-Based Millimeter-Wave Signals for Resilient and Accurate Positioning, Navigation, and Timing
Sponsor: Office of Naval Research (ONR)
Amount: **\$457,000**
Period: 2021–2025
PI: **Z. Kassas**
- **Title:** Aircraft Navigation via Opportunistic Radio Frequency Simultaneous Localization and Mapping
Sponsor: Sandia National Laboratories
Amount: **\$300,000**
Period: 2021–2023
PI: **Z. Kassas**
- **Title:** Assured Navigation and Timing Engineering for Automated Transportation Education and Research – Tier 1 University Transportation Center (UTC)
Sponsor: US Department of Transportation (USDOT)
Amount: \$1,925,000 (Cost Share: \$962,500); **Kassas' Share:** **\$480,966** (Cost Share: **\$190,604**)
Period: 2020–2023
PI: **Z. Kassas**; **Co-PIs:** D. Grejner-Brzezinska (OSU), C. Toth (OSU), K. Redmill (OSU), Ü. Özgüner (OSU), A. O'brien (OSU), Q. Ahmed (OSU), G. Rizzoni (OSU), T. Humphreys (UT-Austin), C. Bhat (UT-Austin), S. Ritchie (UCI), A. Chen (UCI), K. Cohen (UC)
- **Title:** Assessment of Cellular Signals of Opportunity for Aerial Vehicle Navigation
Sponsor: Sandia National Laboratories
Amount: **\$25,000**
Period: 2021
PI: **Z. Kassas**
- **Title:** SNIFFER: Signals of opportunity for Navigation In Frequency-Forbidden EnviRonments
Sponsor: Department of the Air Force
Amount Value: **\$250,000**
Period: 2019–2020
PI: **Z. Kassas**
- **Title:** Developmental Test (DT) Navigation Festival (NAVFEST)
Sponsor: Department of the Air Force

Amount Value: \$43,700

Period: 2019

PI: Z. Kassas

- **Title:** Young Investigator Program (YIP): I Hear, Therefore I Know Where I Am: Exploiting Signals of Opportunity for Robust and Accurate Navigation in GPS-Denied Environments

Sponsor: Office of Naval Research (ONR)

Amount: \$750,000

Period: 2019–2023

PI: Z. Kassas

- **Title:** CAREER: Situational Awareness Strategies for Autonomous Systems in Dynamic Uncertain Environments

Sponsor: National Science Foundation (NSF)

Amount: \$500,000

Period: 2018–2024

PI: Z. Kassas

- **Title:** Ultimate Navigation Chip (uNavChip): Chip-Scale Personal Navigation System Integrating Deterministic Localization and Probabilistic Signals of Opportunity

Sponsor: National Institute of Standards and Technology (NIST)

Amount: \$1,960,613, **Kassas' Share: \$350,000**

Period: 2017–2022

PI: A. Shkel (UCI), **Co-PI:** Z. Kassas, S. Kia (UCI)

- **Title:** Developing a New Generation of Engineers with Cutting-Edge Knowledge in Resilient Navigation and Communication Systems in Contested Electromagnetic Environments

Sponsor: Office of Naval Research (ONR)

Amount: \$750,000

Period: 2016–2022

PI: Z. Kassas

- **Title:** A Collaborative Opportunistic Framework for Resilient and Accurate Navigation in GPS-Challenged Environments

Sponsor: Office of Naval Research (ONR)

Amount: \$510,000

Period: 2016–2020

PI: Z. Kassas

- **Title:** Geophysical Navigation of Unmanned Underwater Vehicles (UUVs)

Sponsor: Office of Naval Research (ONR)

Amount: \$225,000; **Kassas' Share: \$150,000**

Period: 2016–2020

PI: Z. Kassas; Co-PI: J. Farrell (UCR)

- **Title:** Towards Optimal Information Gathering in Unknown Stochastic Environments

Sponsor: National Science Foundation (NSF)

Amount: \$175,000

Period: 2016–2019

PI: Z. Kassas

Internal Grants

- **Title:** Characterization of GPS-Alternative Aircraft Navigation System with Low Earth Orbit (LEO) Satellite Megaconstellations

Sponsor: Consortium on Electromagnetics and Radio Frequencies (CERF), ElectroScience Laboratory (ESL), The Ohio State University

Amount: \$10,000

Period: 2025

PI: Z. Kassas

- **Title:** Characterization of Low Earth Orbit (LEO) Satellite Signals for Positioning and Timing

Sponsor: Consortium on Electromagnetics and Radio Frequencies (CERF), ElectroScience Laboratory (ESL), The Ohio State University

Amount: \$10,000

Period: 2024

PI: Z. Kassas

- **Title:** Characterization of 5G Signals for Positioning, Navigation, and Timing

Sponsor: Consortium on Electromagnetics and Radio Frequencies (CERF), ElectroScience Laboratory (ESL), The Ohio State University

Amount: \$10,000

Period: 2023

PI: Z. Kassas

- **Title:** AI-Driven Intelligent Navigation for Autonomous Vehicles in Challenging Environments

Sponsor: Institute for Information & Communication Technology Promotion (IITP), South Korea

Amount: \$10,000

Period: 2021

PI: Z. Kassas

- **Title:** The Internet of Things (IoT) for Wildfire Management

Sponsor: University of California, Irvine

Amount: \$30,000; **Kassas' Share:** \$12,000

Period: 2020–2021

PI: Lee Swindlehurst (UCI); **Co-PI:** Z. Kassas, H. Jafarkhani (UCI), T. Banerjee (UCI), and A. Eltawil (UCI)

- **Title:** Characterization of cellular LTE signals in a GPS-jammed environment

Sponsor: University of California, Irvine

Amount: \$9,250

Period: 2020–2021

PI: Z. Kassas

**In-Kind
Hardware
& Software
Gifts**

Total: \$276,000

- **Sponsor:** VectorNav
Items: VN-200 Rugged IMU Development Kits
Value: \$14,000
Year: 2024
- **Sponsor:** Orolia
Items: Skydel GNSS Simulation Engine
Value: \$250,000
Year: 2021
- **Sponsor:** Autel Robotics
Items: X-Star Premium Drone Fleet

Value: \$12,000

Year: 2016

Honors & Awards

Outstanding research contributions awards

- Institute of Electrical and Electronics Engineers (IEEE) **Fellow promotion**, 2024
- Institute of Navigation (ION) **Fellow promotion**, 2023
- Ranked by ScholarGPS as the **top scholar globally** in the field of navigation, 2017–2024
- IEEE **Harry Rowe Mimno award**, 2022
- Institute of Navigation (ION) **Colonel Thomas Thurlow award**, 2019
- Institute of Navigation (ION) **Samuel Burka award**, 2018
- IEEE **Senior Member promotion**, 2011

Early career awards

- **Presidential Early Career Award for Scientists and Engineers (PECASE)**, President Biden's White House, 2025
- **Young Investigator Program (YIP) award**, Air Force Office of Scientific Research (AFOSR), 2022
- **Young Investigator Program (YIP) award**, Office of Naval Research (ONR), 2019
- **CAREER award**, National Science Foundation (NSF), 2018
- **Computer and Information Science and Engineering (CISE) Research Initiation Initiative (CRII) award**, NSF, 2016

Plenary, keynote, and distinguished presentations

- **Distinguished lecturer**, IEEE Aerospace and Electronic Systems Society, 2025–2026
- **Distinguished lecturer**, IEEE Intelligent Transportation Systems Society, 2024–2027
- **Keynote speaker**, IEEE International Conference on Acoustics, Speech, & Signal Processing, Workshop on Distributed Signal Processing and Machine Learning for Autonomous Systems, Hyderabad, India, 2025
- **Keynote speaker**, IEEE International Conference on Wireless for Space and Extreme Environments, Workshop on LEO Satellites and Systems, Daytona Beach, FL, 2024
- **Keynote speaker**, IEEE International Symposium on Personal, Indoor and Mobile Radio Communications, Workshop on Mega Constellation Satellite Network for 6G, Toronto, Canada, 2023
- **Distinguished speaker**, 6G Summit, King Abdullah University of Science and Technology (KAUST), Thuwal, Saudi Arabia, 2023
- **Plenary speaker**, International Global Navigation Satellite Systems Conference, Sydney, Australia, 2022
- **Keynote speaker**, IFAC Symposium on Advances in Automotive Control, Columbus, OH, 2022
- **Keynote speaker**, IEEE Vehicular Technology Conference, Workshop on Mission Critical Communications, Helsinki, Finland, 2022
- **Distinguished speaker**, Aerospace Corporation, El Segundo, CA, 2022
- **Keynote speaker**, International Conference on Information Technology: New Gen-

erations, Las Vegas, NV, 2020

- **Keynote speaker**, ION Cognizant Autonomous Systems for Safety Critical Applications Conference, Miami, FL, 2019
- **Plenary speaker**, European Navigation Conference, Gothenburg, Sweden, 2018

Research recognition awards

- **Grand prize**, “An indoor localization system exploiting LTE signals: A synthetic aperture-based beamforming approach to mitigate multipath,” IEEE Signal Processing Society video contest for beamforming research (5-MICC), 2020
- **The University of Texas Research Excellence award**, 2013
- **The University of Texas Professional Development award**, 2011–2013

Best paper awards

- **Best paper award**, “On the fundamental tracking performance and design considerations of radio navigation,” The Ohio State University (OSU) ElectroScience Laboratory (ESL), 2024
- **Best paper award**, “I can hear you loud and clear: GNSS-less aircraft navigation with terrestrial cellular signals of opportunity,” US Department of Transportation Center for Automated Vehicles Research with Multimodal AssurEd Navigation (CARMEN+), 2024
- **Best student paper award**, “Multi-constellation blind beacon estimation, Doppler tracking, and opportunistic positioning with OneWeb, Starlink, Iridium NEXT, and Orbcomm LEO satellites,” IEEE/ION Position, Location, and Navigation Symposium, 2023
- **Best student paper award**, “An interacting multiple model estimator of LEO satellite clocks for improved positioning,” IEEE Vehicular Technology Conference, 2022
- **Best paper award**, “Navigation with differential carrier phase measurements from megaconstellation LEO satellites,” IEEE/ION Position, Location, and Navigation Symposium, 2020
- **Best student paper award**, “Deep learning-aided spatial discrimination for multipath mitigation,” IEEE/ION Position, Location, and Navigation Symposium, 2020
- **IEEE Walter Fried Award for Best Paper**, “Precise UAV navigation with cellular carrier phase measurements,” IEEE/ION Position, Location, and Navigation Symposium, 2018
- **Best student paper award**, “A software-defined receiver architecture for cellular CDMA-based navigation,” IEEE/ION Position, Location, and Navigation Symposium, 2016
- **Best paper award**, “Optimal receiver placement for collaborative mapping of signals of opportunity,” ION GNSS+ Conference, 2015
- **Best student paper award**, “Constructing a continuous phase time history from TDMA signals for opportunistic navigation,” IEEE/ION Position, Location & Navigation Symposium, 2012

Best paper presentation awards

- **Best paper presentation award**, “Improved LEO PNT accuracy enabled by long baseline ephemeris corrections,” ION GNSS+ Conference, 2024
- **Best paper presentation award**, “Improved Starlink satellite orbit prediction via machine learning with application to opportunistic LEO PNT,” ION GNSS+

Conference, 2024

- **Best paper presentation award**, “Blind Doppler tracking and positioning with NOAA LEO satellite signals,” ION GNSS+ Conference, 2023
- **Best paper presentation award**, “Acquisition and tracking of Starlink LEO satellite signals in low SNR regime,” ION GNSS+ Conference, 2023
- **Best research poster award**, “Opportunistic navigation with multi-constellation LEO satellites,” Consortium on Electromagnetics and Radio Frequencies (CERF) Symposium, 2023
- **Best paper presentation award**, “Protecting the skies: GNSS-less aircraft navigation with terrestrial cellular signals of opportunity,” ION GNSS+ Conference, 2022
- **Best paper presentation award**, “Observability analysis of opportunistic receiver localization with LEO satellite pseudorange measurements,” ION GNSS+ Conference, 2022
- **Best paper presentation award**, “Joint detection and tracking of unknown beacons for navigation with 5G signals and beyond,” ION GNSS+ Conference, 2022
- **Best paper presentation award**, “Blind receiver for LEO beacon estimation with application to UAV carrier phase differential navigation,” ION GNSS+ Conference, 2022
- **Best demo award runner-up**, “I am not afraid of the GPS jammer: exploiting cellular signals for accurate ground vehicle navigation in a GPS-denied environment,” ACM Workshop on Automotive and Autonomous Vehicle Security (AutoSec), 2022
- **Best paper presentation award**, “Universal receiver architecture for blind navigation with partially known terrestrial and extraterrestrial signals of opportunity,” ION GNSS+ Conference, 2021
- **Best paper presentation award**, “A machine learning multipath mitigation approach for opportunistic navigation with 5G signals,” ION GNSS+ Conference, 2021
- **Best paper presentation award**, “I am not afraid of the jammer: navigating with signals of opportunity in GPS-denied environments,” ION GNSS+ Conference, 2020
- **Best paper presentation award**, “Blind opportunistic navigation: cognitive deciphering of partially known signals of opportunity,” ION GNSS+ Conference, 2020
- **Best paper presentation award**, “Assessing real 5G signals for opportunistic navigation,” ION GNSS+ Conference, 2020
- **Best paper presentation award**, “Assessment of differential carrier phase measurements from broadband LEO satellite signals for opportunistic navigation,” ION GNSS+ Conference, 2019
- **Best paper presentation award**, “UAV integrity monitoring measure improvement using terrestrial signals of opportunity,” ION GNSS+ Conference, 2019
- **Best paper presentation award**, “Indoor localization based on LTE carrier phase measurements and synthetic antenna array,” ION GNSS+ Conference, 2019
- **Best paper presentation award**, “Centimeter-accurate UAV navigation with cellular signals,” ION GNSS+ Conference, 2018
- **Best paper presentation award**, “Inertial navigation system aiding with Orbcomm LEO satellite Doppler measurements,” ION GNSS+ Conference, 2018
- **Best paper presentation award**, “Distributed signals of opportunity aided inertial navigation with intermittent communication,” ION GNSS+ Conference, 2017
- **Best paper presentation award**, “Computationally efficient receiver design for mitigating multipath for positioning with LTE signals,” ION GNSS+ Conference,

2017

- **Best paper presentation award**, “Signals of opportunity aided inertial navigation,” ION GNSS+ Conference, 2016
- **Best paper presentation award**, “Performance characterization of positioning in LTE systems,” ION GNSS+ Conference, 2016
- **Best research poster finalist**, “Navigation with cellular CDMA signals,” Southern California Robotics Symposium, 2016
- **Best paper presentation award**, “Observability analysis of opportunistic navigation with pseudorange measurements,” AIAA Guidance, Navigation & Control Conference, 2012

Professional service awards

- **IEEE Aerospace and Electronic Systems Society (AESS) outstanding technical panel of the year award**, Navigation systems panel, 2018
- **Institute of Navigation (ION) outstanding peer review recognition award**, 2018

Academic recognition awards

- **Lebanese American University School of Engineering Scholarship**, 2000
- **Lebanese American University School of Engineering Dean’s list of distinguished students**, 1999–2001

Student Supervisee Honors & Awards

- **The Ohio State University (OSU) ElectroScience Laboratory (ESL) Best Ph.D. Dissertation award**, Dr. Mohammad Neinavaie, 2023
- **IEEE Aerospace and Electronic Systems Society (AESS) Robert T. Hill Best Ph.D. Dissertation award**, Dr. Joe Khalife, 2021
- **US Department of Transportation (USDOT) Graduate Student of the Year**, Nadim Khairallah, 2021
- **Melucci Endowed Fellowship**, Mu Jia, 2021
- **Institute of Navigation (ION) Bradford Parkinson award for best Ph.D. Dissertation**, Dr. Kimia Shamaei, 2020
- **Paul and Beverly Holmes Endowed Fellowship**, Ali Abdallah, 2019
- **US Department of Transportation (USDOT) Graduate Student of the Year**, Joshua Morales, 2018
- **University of California, Riverside, Bourns College of Engineering Outstanding Achievement Award**, Souradeep Bhattacharya, 2018
- **University of California, Riverside, Bourns College of Engineering Academic Excellence Award and Dean’s Academic Distinction Award**, Christian Ardito, 2018
- **University of California, Riverside, Graduate Student of the Year**, in the National Center for Sustainable Transportation (NCST) University Transportation Center (UTC), Joe Khalife, 2017
- **35+ Best paper, best student paper, or best paper presentation awards**

Media Visibility

Research featured in dozens of national and international media outlets (Science, BBC, Forbes, The Independent, IEEE Spectrum, ACM, Ars Technica, MIT Technology Review, GPS World, Inside GNSS, Inside Unmanned Systems, Science Magazine, Phys.org, Revolution.aero, Space.com, EurekAlert.org, Homeland Security News, Military Embed-

ded Systems, Cnet.com, Singularity Hub, Yahoo!, Epsilon, EuropaPress, India TV, among others) and appeared on 7 magazine covers

Industrial Experience

National Instruments Corp., Austin, TX **Oct. 2004 – Dec. 2010**
Research and Development (R&D) Engineer

Product Development:

- Developed software: LabVIEW Control Design & Simulation Module, System Identification Toolkit, and PID & Fuzzy Control Toolkit
- Executed various phases (research, implementation, and testing) of the development cycle of six major and minor releases of software
- Coded multi-platform algorithms for off-line design and simulation and Real-Time (RT) and Field Programmable Gate Array (FPGA) deployment
- Led initiatives to improve quality of software, resulting in automated frameworks guaranteeing numerical robustness and reducing testing time from months to hours

Supervision:

- Guided junior engineers in overseas company branches (China and Brazil)
- Served as an industrial advisor for projects from The Ohio State University, Virginia Tech, California State University, and Brigham Young University

Leadership:

- Interviewed candidates for software and hardware R&D positions
- Organized on-campus recruiting booths at Engineering Career Fairs of The Ohio State University
- Assisted with organizing National Instruments Conference (NI Week), Austin, TX, August 2005 & 2007; which attracts 3,000+ engineers, educators, and scientists

Teaching Experience

The Ohio State University **Aug. 2022 – present**

Graduate Courses:

- ECE 5551: State-Space Control Systems (Fall 2023, 2024), 54 students
- ECE 8201: Satellite-Based & Terrestrial Radio Navigation (Fall 2022, 2024), 18 students

University of California, Irvine, CA **Jan. 2019 – Jun. 2022**

Undergraduate Courses:

- MAE 170: Introduction to Control Systems (Winter 2019, 2020, 2022), 165 students

Graduate Courses:

- MAE 295: Global Navigation Satellite System Signal Processing & Software-Defined Radio Design (Fall 2019, 2020), 19 students
- MAE 295: Advanced Detection & Estimation Theory (Spring 2019, 2022), 20 students
- ECPS 209: CPS Case Studies: Applied Optimal Estimation (Winter 2018), 11 students

University of California, Riverside, CA **Oct. 2014 – Dec. 2018**

Graduate Courses:

- EE 235: Linear Systems Theory (Fall 2014, 2015, 2016, 2017, 2018), 105 students
- EE 257: Global Navigation Satellite System Signal Processing & Software-Defined Radio Design (Fall 2015, Spring 2018), 16 students

- EE 260: Advanced Detection & Estimation Theory (Spring 2017), 8 students

Undergraduate Courses:

- EE 132: Automatic Control (Spring 2015, 2016, 2017), 200 students

Texas State University, San Marcos, TX

Jan. – May, 2008 – 2011

Undergraduate Courses:

- MFGE 4376: Control Systems & Instrumentation (Spring 2008, 2009, 2010), 100 students
- EE 4377: Digital Signal Processing (Spring 2011), 13 students

Tutorials

- 2025 LEO PNT: From Theory to Implementation, *IEEE/ION Position, Location, and Navigation Symposium*, Salt Lake City, UT
- 2024 PNT with Terrestrial and LEO Signals of Opportunity, *Consortium on Electromagnetics and Radio Frequencies (CERF)*, The Ohio State University, Columbus, OH

Journal Publications

- [J72] Nasihati, F., & **Kassas, Z.** (2024). Observability analysis of receiver localization with DOA measurements from a single LEO satellite. *IEEE Trans. on Aerospace and Electronic Systems*, submitted.
- [J71] **Kassas, Z.**, & Saroufim, J. (2024). LEO PNT frameworks for non-cooperative satellites with poorly known ephemerides: open-loop SGP4, tracking, and differential. *IEEE Aerospace and Electronic Systems Magazine*, accepted.
- [J70] Saroufim, J., & **Kassas, Z.** (2024). Ephemeris and timing error disambiguation enabling precise LEO PNT. *IEEE Trans. on Aerospace and Electronic Systems*, accepted.
- [J69] Shahcheraghi, S., Saroufim, J., & **Kassas, Z.** (2024). Acquisition, Doppler tracking, and differential LEO-aided IMU navigation with uncooperative satellites. *IEEE Aerospace and Electronic Systems Magazine*, accepted.
- [J68] Hayek, S., & **Kassas, Z.** (2024). Modeling and compensation of timing and spatial ephemeris errors of non-cooperative LEO satellites with application to PNT. *IEEE Trans. on Aerospace and Electronic Systems*, accepted.
- [J67] Kozhaya, S., Saroufim, J., & **Kassas, Z.** (2024). Unveiling Starlink for PNT. *NAVIGATION, Journal of the Institute of Navigation*, (72)1, pp. 1–35.
- [J66] **Kassas, Z.**, Hayek, S., & Haidar-Ahmad, J., (2025). LEO satellite orbit prediction via closed-loop machine learning with application to opportunistic navigation. *IEEE Aerospace and Electronic Systems Magazine*, (40)1, pp. 34–49.
- [J65] **Kassas, Z.**, Neinavaie, M., Khalife, J., Shahcheraghi, S., & Saroufim, J. (2024). The truth is out there: cognitive sensing and opportunistic navigation with unknown terrestrial and nonterrestrial signals. *IEEE Signal Processing Magazine*, (41)5, pp. 87–99 (special issue).
- [J64] **Kassas, Z.**, Abdallah, A., Shahcheraghi, S., Khalife, J., Lee, C., Jurado, J., Wachtel, S., Duede, J., Hoeffner, Z., Hulse, T., Quirarte, R., & Tay, R. (2024). I can hear you loud and clear: GNSS-less aircraft navigation with terrestrial cellular signals of opportunity. *IEEE Trans. on Aerospace and Electronic Systems*, (60)6, pp. 7694–7710.

- [J63] Kozhaya, S., & **Kassas, Z.** (2024). A first look at the OneWeb LEO constellation: beacons, beams, and positioning. *IEEE Trans. on Aerospace and Electronic Systems*, (60)5, pp. 7528–7534.
- [J62] **Kassas, Z.**, Khairallah, N., Khalife, J., Lee, C., Jurado, J., Wachtel, S., Duede, J., Hoeffner, Z., Hulse, T., Quirarte, R., & Tay, R. (2024). Aircraft navigation in GNSS-denied environments via radio SLAM with terrestrial signals of opportunity. *IEEE Trans. on Intelligent Transportation Systems*, (25)10, pp. 14164–14182.
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- [P7] **Kassas, Z.**, Khalife, J., & Abdallah, A. (2019). Receiver design for Doppler positioning with low Earth orbit satellites and differential carrier phase measurements. U.S. Patent Application No. 62/834,317.
- [P6] **Kassas, Z.**, Morales, J., & Khalife, J. (2019). Simultaneous tracking and navigation using LEO satellite signals. U.S. Patent Application No. 62/834,313.
- [P5] **Kassas, Z.**, Maaref, M., & Khalife, J. (2018). Lane-level navigation system for ground vehicles with lidar and cellular signals. U.S. Patent Application No. 62/721,967.
- [P4] **Kassas, Z.**, & Morales, J. (2016). Signals of opportunity aided inertial navigation system. U.S. Patent Application No. 62/398,413.
- [P3] **Kassas, Z.**, Shamaei, K., & Khalife, J. (2016). SDR for navigation with LTE signals. U.S. Patent Application No. 62/398,403.
- [P2] **Kassas, Z.**, Khalife, J., & Shamaei, K. (2016). SDR for navigation with cellular CDMA signals. U.S. Patent Application No. 62/294,758.
- [P1] **Kassas, Z.**, & Lewis, J. (2012). Multi-channel algorithm infrastructure for programmable hardware elements. U.S. Patent No. 8,122,238. Washington, DC: U.S. Patent and Trademark Office.

**Invited
Subject
Matter
Expert**

- 2025 The Pentagon, Washington, DC
- 2024 Hudson Institute, Washington, DC
- 2023 Department of Defense, US Naval Observatory, Washington, DC
- 2022 National Space-Based Positioning, Navigation, and Timing (PNT) Advisory Board, Redondo Beach, CA
- 2022 US Government Accountability Office (GAO), GPS Interference in the Transportation Sector, Virtual
- 2022 US Department of Transportation, Positioning, Navigation, and Timing (PNT) Stakeholder Roundtable, Washington, DC

**Invited
Presentations**

120+ invited presentations at premier academic institutions, international conferences, national laboratories, government agencies, and industry

- 2025 ION Joint Navigation Conference (ION JNC), Covington, KY (*Invited Speaker*)
- 2025 IEEE LEO Satellites and Systems Workshop: LEO Position, Navigating, and Timing (PNT), Virtual (*Invited Speaker*)
- 2025 IEEE/ION Position, Location, and Navigation Symposium, Session: “Frontiers of Radionavigation: Signals of Opportunity, 5G, LEO, and Beyond,” Salt Lake City, UT (*Invited Session*)
- 2025 IEEE/ION Position, Location, and Navigation Symposium, Session: “Frontiers of GNSS,” Salt Lake City, UT (*Invited Session*)
- 2025 George Washington Carver Science Park, Workshop for Research in Low-Earth Orbit, Columbus, OH (*Invited Speaker*)
- 2025 IEEE Aerospace and Electronic Systems Society – Tunisia Chapter, Online Webinar: Ad Astra: Navigation with Megaconstellation LEO Satellites (*IEEE AESS Distinguished Lecture*)

- 2025 IEEE International Conference on Acoustics, Speech, & Signal Processing, Workshop on Distributed Signal Processing and Machine Learning for Autonomous Systems: Synergies of 6G Communications, Learning, and Decision-Making in Multi-Agent Networks, Hyderabad, India (*Keynote Speaker*)
- 2025 Budapest University of Technology and Economics, Budapest, Hungary (*IEEE ITSS Distinguished Lecture*)
- 2025 Department of Information Engineering, University of Padova, Italy (*IEEE AESS Distinguished Lecture*)
- 2025 IEEE Communications Society (ComSoc) Young Professionals (YP), IEEE Women in Communication Engineering (WICE), and IEEE Women in Engineering (WIE) Mentoring Session, Milan, Italy (*Invited Panelist*)
- 2025 IEEE Wireless Communications and Networking Conference, Panel on Advancing Women in STEM and Challenging a Diverse Tomorrow, Milan, Italy (*Invited Panelist*)
- 2024 IEEE International Conference on Wireless for Space and Extreme Environments, Workshop on LEO Satellites and Systems, Daytona Beach, FL (*Keynote Speaker*)
- 2024 IEEE Aerospace and Electronic Systems Society, Online Webinar: Ad Astra: Navigation with Megaconstellation LEO Satellites (*IEEE AESS Distinguished Lecture*)
- 2024 General Robotics, Automation, Sensing, & Perception Lab (GRASP) Lab, University of Pennsylvania, Philadelphia, PA (*IEEE AESS Distinguished Lecture*)
- 2024 US DOT Future of Transportation Summit, Washington, D.C. (*Invited Lectern*)
- 2024 ION Joint Navigation Conference (ION JNC), Covington, KY (*Invited Speaker*)
- 2024 Center for Intelligent Infrastructure, Missouri University of Science and Technology, Rolla, MO (*IEEE AESS Distinguished Lecture*)
- 2024 Department of Electrical Engineering, Chalmers University of Technology, Gothenburg, Sweden (*IEEE AESS Distinguished Lecture*)
- 2024 Department of Electrical and Information Technology, Lund University, Sweden (*IEEE AESS Distinguished Lecture*)
- 2024 Munich Satellite Navigation Summit, Munich, Germany (*Invited Speaker & Panelist*)
- 2024 Munich Aerospace Forum, Munich, Germany (*Invited Speaker*)
- 2024 IEEE Undergraduate Student Chapter, The Ohio State University, Columbus, OH (*Invited Speaker*)
- 2024 Transportation Research Board Annual Meeting, Washington, DC (*Invited Speaker & Panelist*)
- 2023 NSF Workshop on The Convergence of Smart Sensing Systems, Applications, Analytic and Decision Making, Alexandria, VA (*Invited Speaker & Panelist*)
- 2023 Workshop on Space Cybersecurity and Resilience, University of Colorado, Colorado Springs, CO (*Invited Speaker*)
- 2023 IEEE Aerospace and Electronic Systems Society, Online Webinar: No GPS, No Problem: Exploiting Signals of Opportunity for Resilient and Accurate Autonomous Navigation in GPS-Denied Environments (*IEEE AESS Distinguished Lecture*)

- 2023 ION GNSS+ Conference, Session: “Autonomous navigation for ground, seaborne, and airborne vehicles,” Denver, CO (*Invited Session*)
- 2023 Department of Electrical and Computer Engineering, The Ohio State University, Columbus, OH (*IEEE AESS Distinguished Lecture*)
- 2023 IEEE International Symposium on Personal, Indoor and Mobile Radio Communications, Workshop on Mega Constellation Satellite Network for 6G, Toronto, Canada (*Keynote Speaker*)
- 2023 IEEE/ION Position, Location, and Navigation Symposium, Session: “Frontiers of GNSS,” Monterey, CA (*Invited Session*)
- 2023 Department of Mechanical and Aerospace Engineering, West Virginia University, Morgantown, WV (*IEEE AESS Distinguished Lecture*)
- 2023 Department of Aerospace Engineering & Mechanics, University of Minnesota, Twin Cities, Minneapolis, MN (*IEEE AESS Distinguished Lecture*)
- 2023 6G Summit, King Abdullah University of Science and Technology (KAUST), Thuwal, Saudi Arabia (*Distinguished Speaker*)
- 2022 International Global Navigation Satellite Systems Conference, Sydney, Australia (*Plenary Speaker*)
- 2022 International Global Navigation Satellite Systems Conference, Session: “PNT in Low Earth Orbit,” Sydney, Australia (*Panelist*)
- 2022 International Global Navigation Satellite Systems Conference, Session: “5G,” Sydney, Australia (*Panelist*)
- 2022 IEEE Low Earth Orbit (LEO) Satellites & Systems Workshop, Virtual (*Invited Speaker*)
- 2022 ITS America, Webinar: “The New Infrastructure: Physical + Digital Helps Enable Positive Citizen Outcomes,” (*Invited Speaker & Panelist*)
- 2022 ION GNSS+ Conference, Session: “Autonomous Navigation for Ground, Seaborne, and Airborne Vehicles,” Denver, CO (*Invited Speaker & Panelist*)
- 2022 ION GNSS+ Conference, Session: “Beyond GNSS: Emerging Trends in LEO-Based and Terrestrial Signals of Opportunity for PNT,” Denver, CO (*Invited Speaker & Panelist*)
- 2022 IFAC Symposium on Advances in Automotive Control, Columbus, OH (*Keynote Speaker*)
- 2022 GPS World Magazine, Online Webinar: “A Galaxy Far, Far Away? Why LEO PNT is Closer to Reality Than You Think” (*Invited Speaker*)
- 2022 IEEE Vehicular Technology Conference, Workshop on Mission Critical Communications, Helsinki, Finland (*Keynote Speaker*)
- 2022 Aerospace Corporation, El Segundo, CA (*Distinguished Speaker*)
- 2022 Munich Satellite Navigation Summit, Munich, Germany (*Invited Speaker & Panelist*)
- 2021 GPS World Magazine, Online Webinar: “Measuring PNT Performance in Challenged Environments” (*Invited Speaker*)
- 2021 Department of Mechanical Engineering, Tufts University, Medford, MA (*Seminar*)
- 2021 ION GNSS+ Conference, Session: “AI-Enabled Navigation in Smart Cities,” St. Louis, MO (*Invited Speaker*)
- 2021 ION GNSS+ Conference, Session: “On the Road to Automated Vehicles,” St. Louis, MO (*Invited Speaker & Panelist*)
- 2021 Lebanese American University, Byblos, Lebanon (*Seminar*)

- 2021 National Institute of Standards and Technology (NIST), Public Safety Broadband Stakeholder Meeting, Virtual (*Invited Speaker*)
- 2021 International Technical Symposium on Navigation and Timing, Toulouse, France (*Invited Speaker*)
- 2021 ASCE International Conference on Transportation and Development, Virtual (*Invited Speaker & Panelist*)
- 2021 Safety and Security for Connected Autonomous Vehicles, University of California, Irvine, Virtual (*Invited Speaker & Panelist*)
- 2021 Air Force Research Laboratory, Wright-Patterson Air Force Base, OH, Virtual (*Seminar*)
- 2021 US–EU GPS–Galileo Working Group C, Resilience Subgroup, Virtual (*Invited Speaker*)
- 2020 Workshop on Spectrum Challenges and Opportunities for 6G and Navigation, Stanford University, Stanford, CA (*Invited Speaker & Panelist*)
- 2020 ION GNSS+ Conference, Session: “PNT for AI-Enabled Autonomous Systems: The Good, the Bad, and the Ugly,” St. Louis, MO (*Invited Speaker & Panelist*)
- 2020 National Institute of Standards and Technology (NIST), Public Safety Broadband Stakeholder Meeting, Virtual (*Invited Speaker*)
- 2020 International Conference on Information Technology: New Generations, Las Vegas, NV (*Keynote Speaker*)
- 2020 University of California, Irvine, CA (*Invited Speaker & Panelist*)
- 2020 Information Theory and Applications Workshop, University of California, San Diego, CA (*Invited Speaker*)
- 2019 Institute of Navigation (ION), Online Webinar: “LTE Receiver Design and Multipath Analysis for Navigation in Urban Environments” (*Invited Speaker*)
- 2019 IEEE Vehicular Technology Conference, Honolulu, HI (*Invited Speaker & Panelist*)
- 2019 ION GNSS+ Conference, Session: “The Spectrum of Autonomy in Navigation,” Miami, FL (*Invited Speaker & Panelist*)
- 2019 ION GNSS+ Conference, Session: “PNT Security and Robustness,” Miami, FL (*Invited Speaker & Panelist*)
- 2019 ION Cognizant Autonomous Systems for Safety Critical Applications Conference, Miami, FL (*Keynote Speaker*)
- 2019 National Institute of Standards and Technology (NIST), Public Safety Broadband Stakeholder Meeting, Chicago, IL (*Invited Speaker*)
- 2019 Naval Surface Warfare Center, Corona, CA (*Seminar*)
- 2019 University of California, Irvine, CA (*Invited Speaker & Panelist*)
- 2019 École de Technologie Supérieure (ÉTS), Montreal, Canada (*Seminar*)
- 2018 Asilomar Conference on Signals, Systems and Computers, Session: “Signal Processing for GNSS and/or Localization with Terrestrial Networks,” Pacific Grove, CA (*Invited Session*)
- 2018 ION GNSS+ Conference, Session: “Autonomous Cyber-Physical Systems – The Way Ahead,” Miami, FL (*Invited Speaker*)
- 2018 Naval Surface Warfare Center, Corona, CA (*Seminar*)
- 2018 IFAC Networked & Autonomous Air & Space Systems Workshop, Session: “UAV Estimation & Control,” Santa Fe, NM (*Invited Speaker in Plenary Session*)

- 2018 European Navigation Conference, Gothenburg, Sweden (*Plenary Speaker*)
- 2018 IEEE Wireless Communications and Networking Conference, Session: “Localization in Current and Emerging Networks,” Barcelona, Spain (*Invited Session*)
- 2018 Department of Aeronautics & Astronautics, University of Washington, Seattle, WA (*Seminar*)
- 2018 Department of Aerospace Engineering Sciences, University of Colorado Boulder, CO (*Seminar*)
- 2018 Department of Electrical & Computer Engineering, University of California, Santa Barbara, CA (*Seminar*)
- 2018 ION Cognizant Autonomous Systems for Safety Critical Applications Workshop, Reston, VA (*Keynote Speaker*)
- 2017 Department of Mechanical & Aerospace Engineering, University of California, San Diego, CA (*Seminar*)
- 2017 Position, Navigation, & Time (PNT) Symposium, Stanford University, Stanford, CA (*Invited Speaker*)
- 2017 Texas Wireless Summit, Wireless Networking & Communications Group (WNCG), The University of Texas at Austin, TX (*Invited Speaker & Panelist*)
- 2017 Department of Aerospace Engineering & Engineering Mechanics, The University of Texas at Austin, TX (*Seminar*)
- 2017 Naval Surface Warfare Center, Corona, CA (*Seminar*)
- 2017 European Signal Processing Conference, Session: “Positioning In Challenging Environments,” Kos Island, Greece (*Invited Session*)
- 2017 GPS World Magazine, Online Webinar: “UAVs: From Flying Drones to Doing Business” (*Invited Speaker*)
- 2017 Department of Electrical & Computer Engineering, University of California, Riverside, CA (*Seminar*)
- 2017 Department of Mechanical, Materials & Aerospace Engineering, Illinois Institute of Technology, Chicago, IL (*Seminar*)
- 2017 Department of Electrical Engineering, University of California, Los Angeles, CA (*Seminar*)
- 2017 Department of Electrical & Computer Engineering, The Ohio State University, Columbus, OH (*Seminar*)
- 2017 Department of Electrical & Computer Engineering, Carnegie Mellon University, Silicon Valley, CA (*Seminar*)
- 2017 Department of Electrical Engineering & Computer Science, University of California, Irvine, CA (*Seminar*)
- 2016 Society of Women Engineers, University of California, Riverside, CA (*Invited Speaker*)
- 2016 Department of Electrical Engineering & Computer Science, Ohio University, Athens, OH (*Seminar*)
- 2016 Department of Electrical & Computer Engineering, Air Force Institute of Technology, Wright-Patterson Air Force Base, OH (*Seminar*)
- 2016 Department of Electrical & Computer Engineering, The Ohio State University, Columbus, OH (*Seminar*)
- 2016 Department of Aeronautics & Astronautics, University of Washington, Seattle, WA (*Seminar*)
- 2016 ION Southern California Section, John Deere, Torrance, CA (*Seminar*)

- 2016 Department of Aeronautics & Astronautics, Stanford University, Stanford, CA (*Seminar*)
- 2015 Department of Electrical Engineering, University of South Florida, Tampa, FL (*Seminar*)
- 2015 Department of Electrical & Computer Engineering, University of California, Santa Barbara, CA (*Seminar*)
- 2015 Department of Electrical & Computer Engineering, University of California, Riverside, CA (*Seminar*)
- 2015 Naval Research Laboratory, Washington, D.C. (*Seminar*)
- 2014 Department of Aerospace Engineering & Mechanics, University of Minnesota, Twin Cities, Minneapolis, MN (*Seminar*)
- 2014 Department of Electrical & Computer Engineering, American University of Beirut, Lebanon (*Seminar*)
- 2014 Department of Electrical & Computer Engineering, University of California, Riverside, CA (*Seminar*)
- 2013 IEEE Global Conference on Signal & Information Processing, Session: “Software Defined and Cognitive Radios,” Austin, TX (*Invited Session*)
- 2013 The Aerospace Corporation, El Segundo, CA (*Seminar*)
- 2013 Northrop Grumman Electronic Systems, Woodland Hills, CA (*Seminar*)
- 2013 Charles Stark Draper Laboratory, Cambridge, MA (*Seminar*)
- 2013 Robotics & Autonomous Vehicles Summit, National Instruments Week Conference, Austin, TX (*Invited Speaker*)
- 2013 Academic Forum, National Instruments Week Conference, Austin, TX (*Invited Speaker*)
- 2012 The University of Texas at Austin, Electrical & Computer Engineering Graduate Lecture Series, Austin, TX (*Invited Panelist*),
- 2012 Texas Wireless Summit, Austin, TX (*Invited Session*)
- 2012 National Science Foundation Wireless Internet Center for Advanced Technology (NSF-WICAT), Virginia Tech, Blacksburg, VA (*Invited Session*)
- 2008 Department of Electrical & Computer Engineering, American University of Beirut, Lebanon (*Seminar*)
- 2007 Mechatronics Symposium, National Instruments Week Conference, Austin, TX (*Invited Speaker*)
- 2007 Air Force Research Laboratory, Wright-Patterson Air Force Base, OH (*Seminar*)
- 2007 L-3 Communications, Cincinnati, OH (*Seminar*)
- 2007 Belcan Engineering, Cincinnati, OH (*Seminar*)
- 2007 Center for Automotive Research, The Ohio State University, Columbus, OH (*Seminar*)
- 2005 Department of Electrical & Computer Engineering, The Ohio State University, Columbus, OH (*Seminar*)

Committees and Panels

- **Chair**, *IEEE Aerospace and Electronic Systems Society, Navigation Systems Panel* (2023 – present)
- **Fellows evaluation committee**, *IEEE Aerospace and Electronic Systems Society* (2025)
- **Selection committee**, *IEEE AESS Awards* (2025)
- **Panelist**, *IEEE Vehicular Technology Society, Committee on Mission Critical Com-*

munications (2021 – present)

- **Panelist**, *IEEE Aerospace and Electronic Systems Society, Vision and Perspectives Panel* (2021 – 2024)
- **Selection committee**, *ION Bradford Parkinson Award for Best Ph.D. Dissertation* (2022, 2018)
- **Selection committee**, *ION GNSS+ Conference Best Student Paper Award* (2022, 2021)
- **Vice chair**, *IEEE Aerospace and Electronic Systems Society, Navigation Systems Panel* (2021 – 2023)
- **Panelist**, *IEEE Aerospace and Electronic Systems Society, Navigation Systems Panel* (2017 – 2021)
- **Council member-at-large**, *Institute of Navigation (ION)* (2019 – 2021)
- **Selection committee**, *IEEE Senior Membership Advancement*, Austin, TX (2013)

Review Panels

- 2025 **Review panelist**, *National Science Foundation (NSF), Secure and Trustworthy Cyberspace (SaTC)*, Alexandria, VA
- 2025 **Review panelist**, *National Science Foundation (NSF), Cyber-Physical Systems (CPS)*, Alexandria, VA
- 2024 **Review panelist**, *National Science Foundation (NSF), Computer and Network Systems (CNS)*, Alexandria, VA
- 2023 **Review panelist**, *National Science Foundation (NSF), Cyber-Physical Systems (CPS)*, Alexandria, VA
- 2022 **Review panelist**, *National Science Foundation (NSF), Cyber-Physical Systems (CPS)*, Alexandria, VA
- 2022 **Review panelist**, *National Science Foundation (NSF), Computer and Network Systems (CNS)*, Alexandria, VA
- 2021 **Review panelist**, *National Science Foundation (NSF), Smart and Connected Communities (S²CC)*, Alexandria, VA
- 2021 **Review panelist**, *National Science Foundation (NSF), Cyber-Physical Systems (CPS)*, Alexandria, VA
- 2019 **Review panelist**, *National Science Foundation (NSF), Cyber-Physical Systems (CPS)*, Alexandria, VA
- 2019 **Review panelist**, *National Science Foundation (NSF), Secure and Trustworthy Cyberspace (SaTC)*, Alexandria, VA
- 2019 **Review panelist**, *National Science Foundation (NSF), Cyber-Physical Systems (CPS)*, Alexandria, VA
- 2018 **Review panelist**, *National Science Foundation (NSF), Smart and Autonomous Systems (S²AS)*, Alexandria, VA
- 2018 **Review panelist**, *National Science Foundation (NSF), Cyber-Physical Systems (CPS)*, Alexandria, VA
- 2018 **Review panelist**, *National Science Foundation (NSF), Computer and Network Systems (CNS)*, Alexandria, VA
- 2018 **Review panelist**, *National Science Foundation (NSF), Cyber-Physical Systems (CPS)*, Alexandria, VA
- 2017 **Review panelist**, *National Science Foundation (NSF), Cyber-Physical Systems (CPS)*, Alexandria, VA

**Conference
Organization**

- 2017 **Review panelist**, *National Science Foundation (NSF), Computer and Network Systems (CNS)*, Arlington, VA
- 2017 **Review panelist**, *National Science Foundation (NSF), Cyber-Physical Systems (CPS)*, Arlington, VA
- 2017 **Review panelist**, *National Science Foundation (NSF), Smart and Autonomous Systems (SEAS)*, Arlington, VA
- 2016 **Review panelist**, *National Science Foundation (NSF), Computer and Network Systems (CNS)*, Arlington, VA
- 2016 **Review panelist**, *National Science Foundation (NSF), Cyber-Physical Systems (CPS)*, Arlington, VA
- 2025 **Co-organizer and co-chair**, *ION GNSS+ Conference*, “Panel: LEO Constellations for PNT: Services, Opportunities, and Challenges,” Baltimore, MD
- 2025 **Co-organizer and co-chair**, *IEEE Vehicular Technology Conference*, “Workshop: Mission Critical Communications,” Oslo, Norway
- 2025 **Track chair**, *IEEE/ION Position, Location and Navigation Symposium*, “Integrated and Opportunistic Navigation,” , Salt Lake City, CA
- 2024 **Co-organizer and co-chair**, *IEEE International Conference on Wireless in Space and Extreme Environments*, “Workshop: IEEE Low-Earth Orbit (LEO) Satellites and Systems (SatS),” Daytona Beach, FL
- 2024 **Co-organizer**, *International Conference on Information Fusion*, “Special Session: Multi-modal Fusion for Assured Positioning, Navigation, and Timing (PNT),” Venice, Italy
- 2023 **Co-organizer and co-chair**, *ION GNSS+ Conference*, “Panel: Autonomous Navigation for Ground, Seaborne, and Airborne Vehicles,” Denver, CO
- 2023 **Co-organizer and co-chair**, *IEEE Vehicular Technology Conference*, “Workshop: Mission Critical Communications,” Florence, Italy
- 2023 **Program chair**, *IEEE/ION Position, Location and Navigation Symposium*, Monterey, CA
- 2022 **Co-organizer**, *International Conference on Information Fusion*, “Special Session: Estimation and Fusion for Navigation,” Linköping, Sweden
- 2022 **Co-organizer and co-chair**, *IEEE Vehicular Technology Conference*, “Workshop: Mission Critical Communications,” Helsinki, Finland
- 2022 **Organizer**, *Munich Satellite Navigation Summit*, “Special Session: AI and Advanced Technologies for Next Generation GNSS,” Munich, Germany
- 2021 **Track chair**, *ION GNSS+ Conference*, “Algorithms and Methods,” Saint Louis, MO
- 2021 **Co-organizer and co-chair**, *ION GNSS+ Conference*, “Panel: AI-Enabled Navigation in Smart Cities,” Saint Louis, MO
- 2021 **Co-organizer and co-chair**, *IEEE International Conference on Intelligent Transportation Systems*, “Special Session: Resilient and Secure Navigation for Highly Automated Transportation Systems,” Indianapolis, IN
- 2020 **Organizer and chair**, *ION GNSS+ Conference*, “Panel: PNT for AI-Enabled Autonomous Systems: The Good, the Bad, and the Ugly,” Saint Louis, MO
- 2020 **Program chair**, *IEEE/ION Position, Location and Navigation Symposium*, Portland, OR

	2020	Organizer and chair , <i>IEEE International Conference on Acoustics, Speech, and Signal Processing</i> , “Special Session: Signal Processing for Sensing, Information Fusion, and Situational Awareness in Autonomous Systems,” Barcelona, Spain
	2019	Co-organizer and co-chair , <i>IEEE Vehicular Technology Conference</i> , “Workshop: Reliable Ubiquitous Navigation in Smart Cities,” Honolulu, HI
	2019	Co-organizer and co-chair , <i>ION GNSS+ Conference</i> , “Panel: Navigating Smart and Connected Cities,” Miami, FL
	2019	General chair , <i>ION Cognizant Autonomous Systems for Safety Critical Applications Conference</i> , Miami, FL
	2018	Co-organizer and co-chair , <i>ION GNSS+ Conference</i> , “Panel: Autonomous Cyber-Physical Systems – The Way Ahead,” Miami, FL
	2018	Track chair , <i>IEEE/ION Position, Location and Navigation Symposium</i> , “Applications to Automated, Semi-Autonomous, and Fully-Autonomous Systems,” Monterey, CA
	2018	Organizer and general chair , <i>ION Cognizant Autonomous Systems for Safety Critical Applications Workshop</i> , Reston, VA
Professional Service	2025	Session co-chair , “Beyond GNSS: Emerging Trends in LEO-Based and Terrestrial Signals of Opportunity for PNT,” <i>ION GNSS+ Conference</i> , Baltimore, MD
	2025	Technical program committee member , <i>IEEE Vehicular Technology Conference</i> , Oslo, Norway
	2024	Committee member , “Reducing Cybersecurity Risks,” <i>US DOT Future of Transportation Summit</i> , Washington, D.C.
	2023	Session chair , “Localization and Direction Finding,” <i>IEEE Vehicular Technology Conference</i> , Florence, Italy
	2023	Technical program committee member , <i>IEEE Vehicular Technology Conference</i> , Florence, Italy
	2022	Session co-chair , “Indoor Navigation and Positioning,” <i>ION GNSS+ Conference</i> , Denver, CO
	2022	Session co-chair , “Position, Navigation and Timing Security in Highly Automated Vehicles,” <i>IFAC Symposium on Advances in Automotive Control</i> , Columbus, OH
	2022	Session chair , “Positioning 1,” <i>IEEE Vehicular Technology Conference</i> , Helsinki, Finland
	2022	International program committee member , <i>IFAC Symposium on Advances in Automotive Control</i> , Columbus, OH
	2022	Technical program committee member , <i>IEEE Vehicular Technology Conference</i> , Helsinki, Finland
	2022	Session chair , “Autonomous Driving Security 2,” <i>ACM Workshop on Automotive and Autonomous Vehicle Security</i> , San Diego, CA
	2022	Session co-chair , “Radionavigation with Terrestrial and LEO Signals,” <i>ION International Technical Meeting</i> , Long Beach, CA
	2021	Technical program committee member , <i>IEEE Military Communications Conference</i> , San Diego, CA
	2021	Associate editor , <i>American Control Conference</i> , New Orleans, LA

- 2021 **Technical program committee member**, *IEEE Vehicular Technology Conference*, Helsinki, Finland
- 2021 **Session co-chair**, “Radionavigation Beyond Medium Earth Orbit GNSS,” *ION International Technical Meeting*, San Diego, CA
- 2020 **Technical program committee member**, *IEEE Vehicular Technology Conference*, Victoria, Canada
- 2019 **Session chair**, “Aerial Vehicle Autonomy 2,” *ION Cognizant Autonomous Systems for Safety Critical Applications Conference*, Miami, FL
- 2019 **Session chair**, “Localization Techniques,” *IEEE Vehicular Technology Conference*, Kuala Lumpur, Malaysia
- 2018 **Session chair**, “Active Safety and Autonomous Driving,” *European Navigation Conference*, Gothenburg, Sweden
- 2018 **Session co-chair**, “Autonomous Navigation,” *ION International Technical Meeting*, Reston, VA
- 2017 **Session co-chair**, “Multisensor Navigation in Challenging Environments,” *ION GNSS+ Conference*, Portland, OR
- 2017 **Session co-chair**, “Autonomous Robots II,” *American Control Conference*, Seattle, WA
- 2017 **Associate editor**, *American Control Conference*, Seattle, WA
- 2017 **Session co-chair**, “Autonomous Navigation,” *ION International Technical Meeting*, Monterey, CA
- 2016 **Session co-chair**, “Collaborative and Networked Navigation,” *IEEE/ION Position, Location and Navigation Symposium*, Savannah, GA
- 2016 **Session co-chair**, “Multi-Sensor Fusion,” *ION International Technical Meeting*, Monterey, CA
- 2013 **Session co-chair**, “Sensor Systems for GNC II,” *AIAA Guidance, Navigation, & Control Conference*, Boston, MA
- 2011 **Session co-chair**, “Control Applications,” *IFAC World Congress*, Milan, Italy

Journal Editorship

- **Senior editor**, *IEEE Transactions on Intelligent Transportation Systems*, (2023 – present).
- **Associate editor**, *IEEE Transactions on Aerospace and Electronic Systems*, (2016 – present).
- **Senior editor**, *IEEE Transactions on Intelligent Vehicles*, (2021 – 2023).
- **Associate editor**, *IEEE Transactions on Intelligent Transportation Systems*, (2019 – 2023).
- **Guest editor**, *IEEE Intelligent Transportation Systems Magazine, Special issue on Recent Advances in the Use of GNSS-based Positioning for Intelligent Transport Systems*, (2018 – 2019).

Peer-Review Activities

Journals

- IEEE Transactions on Aerospace and Electronic Systems
- IEEE Transactions on Intelligent Transportation Systems
- IEEE Transactions on Intelligent Vehicles
- IEEE Transactions on Wireless Communications
- IEEE Transactions on Signal Processing
- IEEE Journal of Selected Topics in Signal Processing

- IEEE Journal on Selected Areas in Communications
- IEEE Aerospace and Electronic Systems Magazine
- IEEE Intelligent Transportation Systems Magazine
- IEEE Vehicular Technology Magazine
- NAVIGATION, Journal of the Institute of Navigation
- IEEE Control Systems Letters
- SPIE Journal of Applied Remote Sensing

Conferences

- IEEE/ION Position, Location, and Navigation Symposium
- IEEE Vehicular Technology Conference
- ION GNSS+ Conference
- IEEE Intelligent Vehicles Symposium
- IEEE International Conference on Acoustics, Speech, and Signal Processing
- IEEE Military Communications Conference
- IEEE Wireless Communications and Networking Conference
- ION International Technical Meeting
- IEEE Conference on Control Technology and Applications
- European Signal Processing Conference
- AIAA Guidance, Navigation, and Control Conference
- American Control Conference
- IEEE Multiconference on Systems and Control
- IFAC World Congress

Professional Memberships

- Institute of Electrical and Electronics Engineers (IEEE), Fellow
- Institute of Navigation (ION), Fellow
- IEEE Aerospace and Electronic Systems Society, IEEE Intelligent Transportation Systems Society, IEEE Vehicular Technology Society, IEEE Signal Processing Society, IEEE Communications Society

University Committees

The Ohio State University

Department of Electrical & Computer Engineering

- Area Chair (Systems: Control, Signal Processing, & Computer Vision), 2024 – present
- Promotion & Tenure Committee, 2023 – present
- Curriculum Committee, 2024 – present
- Recruiting & Financial Aid Committee, 2023 – 2024
- Personnel Committee, 2022 – 2023

ElectroScience Laboratory

- Personnel Committee, 2022 – present

College of Engineering

- Research Committee, 2023 – present

University of California, Irvine

Department of Mechanical & Aerospace Engineering

- Faculty Search Committee, 2019 – 2020

- Graduate Studies Committee, 2018 – 2019

Campus

- Campus Selection Committee for proposal submission to the Department of Defense Research and Education Program (REP) for Historically Black Colleges and Universities/Minority Serving Institutions (HBCU/MI), 2021

University of California, Riverside

Campus

- Cluster Hiring in Autonomy, Robotics, and Dynamics Committee, 2016

Bourns College of Engineering

- Research & Instructional Infrastructure Committee, 2016

Department of Electrical & Computer Engineering

- Colloquium Series Co-organizer (2016 – 2017, 2017 – 2018)
- Undergraduate Curriculum Committee, 2016
- Outreach, Recruitment, & Publicity Committee, 2015
- Department Webpage Redesign Committee, 2015

Visiting Faculty Hosted

- Prof. Laura Ruotsalainen (University of Helsinki, Finland), 2023
- Prof. Jiwon Seo (Yonsei University, South Korea), 2020 – 2021

Postdoctoral Researcher Supervision

- Dr. Forough Nasihati, 2023 – 2024 (*Job upon conclusion: The Ohio State University*)
- Dr. Zeinab Shadram, 2021 – 2022 (*Job upon conclusion: PHY Wireless*)
- Dr. Joe Khalife, 2020 – 2021 (*Job upon conclusion: Apple*)
- Dr. Mahdi Maaref, 2017 – 2020 (*Job upon conclusion: oneNav*)

Ph.D. Student Supervision

- Jennifer Sanderson
- Paul El Kouba
- Faezeh Mooseli
- Artun Sel
- Joe Saroufim
- Samer Watchi Hayek
- Shaghayegh Shahcheraghi
- Sharbel Kozhaya
- Dr. Mohammad Neinavaie, Dissertation: “Cognitive sensing and navigation with unknown terrestrial and LEO satellite signals,” 2023 (*Job upon graduation: Apple*)
- Dr. Kimia Shamaei, Dissertation: “Exploiting cellular signals for navigation: 4G to 5G,” 2020 (*Job upon graduation: Apple*)
- Dr. Joe Khalife, Dissertation: “Precise navigation with cellular signals: Receiver design, differential and non-differential frameworks, and performance analysis,” 2020 (*Job upon graduation: Postdoctoral researcher, University of California, Irvine*)
- Dr. Joshua Morales, Dissertation: “Inertial radio SLAM: Standalone and collaborative architectures,” 2020 (*Job upon graduation: StarNav*)

Visiting Ph.D. Student Supervision	• Halim Lee (Yonsei University, South Korea), 2020 – 2021
M.S. Student Supervision	• Will Barrett • Nadim Khairallah, Thesis: “Towards opportunistic navigation with LEO satellites: Adaptive clock estimation and closed-loop ephemeris tracking,” 2022 (<i>Job upon graduation: SpaceX</i>) • Alexander Nguyen, Thesis: “Aerial vehicle navigation with terrestrial signals of opportunity: Performance analysis and transmitter selection,” 2022 (<i>Job upon graduation: Air Force Research Laboratory</i>) • Ali Abdallah, Thesis: “Indoor navigation with cellular signals utilizing synthetic aperture for multipath mitigation,” 2022 (<i>Job upon graduation: Google</i>) • Sonya Ragothaman, Thesis: “Path Planning for Autonomous Ground Vehicles using GNSS and Cellular LTE Signal Reliability Maps and GIS 3-D Maps,” 2018 (<i>Job upon graduation: The Aerospace Corporation</i>) • Jesse Garcia, Thesis: “Autonomous Surface Vehicle Measurement Location Planning for Optimal Underwater Acoustic Transponder Localization,” 2018 (<i>Job upon graduation: Naval Information Warfare Center Pacific, San Diego</i>) • Ralph Sabbagh, non-thesis, 2023 (<i>Job upon graduation: Ph.D. student, University of California, Irvine</i>) • Jamil Haidar Ahmad, non-thesis, 2022 (<i>Job upon graduation: ASML</i>) • Mohamad Orabi, non-thesis, 2022 (<i>Job upon graduation: oneNav</i>) • Trier Mortlock, non-thesis, 2021 (<i>Job upon graduation: Ph.D. student, University of California, Irvine</i>) • Christian Ardito, non-thesis, 2020 (<i>Job upon graduation: MaXentric Technologies</i>)
B.S. Thesis Supervision	• Carlos Acebes Cebrián, Double B.S. Thesis in Aerospace Systems Engineering and Telecommunication Systems Engineering: “Opportunistic Navigation with Iridium Next LEO Satellites,” 2020 (<i>Job upon graduation: Indra Sistemas</i>)
Undergraduate Student Researcher Supervision	[30] Jonathan Low (Jan. 2022 – Mar. 2022) [29] Shiming Xu (Sep. 2021 – Mar. 2022), [28] Eddie Huang (Jan. 2020 – Mar. 2022), [27] Oleon Hariharan (Apr. 2021 – Dec. 2021), [26] Vivek Hatte (Apr. 2021 – Jun. 2021), [25] Giles Pierre Nunez Carlos (Apr. 2021 – Jun. 2021), [24] Raphael Feliciano (Oct. 2020 – Jun. 2021), [23] Andrew Zhao (Oct. 2020 – Jun. 2021), [22] Sharbel Kozhaya (Jul. 2020 – Aug. 2020), [21] Qitai (Neo) Meng (Oct. 2019 – Jun. 2020), [20] Yanhao Yang (Oct. 2019 – Jun. 2020), [19] Xinyi Taylor Zhang (Jun. 2019 – Mar. 2020), [18] Zainab Ashai (Apr. 2019 – Mar. 2020), [17] Bhavani Panda (Apr. 2019 – Dec. 2019), [16] Aishwarya Bhandari (Apr. 2019 – Dec. 2019), [15] Michael Maher (Oct. 2019 – Dec. 2019), [14] MyLinh Nguyen (Apr. 2019 – Sep. 2019), [13] Labiba Saleh (Jun. 2019 – Sep. 2019), [12] Mohamad Orabi (Jun. 2019 – Aug. 2019), [11] Naji Tarabay (Jun. 2019 – Aug. 2019), [10] Brandon Lam (Apr. 2019 – Jun. 2019), [9] Jerry Lee (Jan. 2019 – Jun. 2019), [8] Kyle Semelka (Apr. 2018 – Jun. 2019), [7] Chester Lau (Jul. 2018 – Apr. 2019), [6] Souradeep Gogol Bhattacharya (Oct. 2015 – Jun. 2018), [5] Christian Ardito (Mar. 2018 – Jun. 2018), [4] Sonya Ragothaman (Apr. 2016 – Sep. 2016), [3] Jesse Garcia (Apr. 2016 – Sep. 2016), [2] Gustavo Correa (Jul. 2017 – Dec. 2017), [1] Farah Khalife (Jun. 2015 – Aug. 2015)

**High School
Student
Supervision**

[2] Frank Li (Jun. 2019 – Aug. 2019), [1] Shintaro Mori (Jun. 2017 – Jun. 2018)

**Senior Design
Project**

Co-supervision

- “Navigating the Skies with Starlink,” Will Barrett, Jackson Morris, Ethan Shaw, Vasilios Konstantacos, May 2024
- “UAV Sensing and Navigation with Unknown Terrestrial Transmitters,” Khaled Hamil, Aiden Short, Andrew Ye, May 2024
- “CubeSat,” Taekyoo Won, Elias Andraos, Andraos Arain, Sarah Catania Orozco, Edwin Christhuraj, Manasi Deshpande, Adam Adam, Aileen Juang, Jimmy Juarez, Tasmima Khan, Tasmima Koo, Tyler Lee, Bryan Matel, Krish Mehta, Shreejan Mistry, Adrian Osorio, Gabrielle Palar, Gabrielle Phan, Kendrick Phan, Kendrick Prata, Sherlock Qin, Erik Ren, Diego Rodriguez Orozco, Brianna Sandoval, Alexandria Shin, Enson Su, Raineir Tabano, Yi-Ju Tsai, Gary Villar, Grazelle Vilorio, Duong Vu, April Wu, Mengjie Xie, Catherine Zhang, Jun. 2022
- “Unmanned Floating Vehicle,” Eli Tsao, Chaz Fazio, Reece Rivera, Wessam Elmasri, Mar. 2022
- “Autonomous Target Robot,” Erik Sandelin, Huy Ho, Sary Aranki, Megan Uozumi, Anthony Atz, Mar. 2022
- “CubeSat,” Taekyoo Won, Suleyman Varlibas, Piunik Haghnazarian, Kelby Custodio, James Bohne, Anthony Castillo, Mohamed Abdelwahab, Angel Alarcon, Alarcon Chen, Edwin Christhuraj, Sarah Catania Orozco, Tyler Cook, An Thanh Dang, Andrew Kettle, Jaeven Laron, Tyler Lee, Alexei Lorion, Adrian Osorio, Elijah Reed, Jason Sanchez, Sergio Sandoval, Alexandria Shin, Arianne Agno, Hafsah Arain, Oscar Castro, Manasi Deshpande, Aileen Juang, Xavier Lian, Weixin Lin, Andrew Nguyen, Aditi Pai, Micah Raney, Erik Ren, Diego Rodriguez Orozco, Brianna Sandoval, Enson Su, Adel Tani, Junyang Zhang, Jun. 2021.
- “CubeSat,” Taekyoo Won, Armen Ter Avetisyan, Sergio Sandoval, Baldwin Ngo, Edwin Christhuraj, Syed Hassan, Akash Idnani, Jaeven Laron, Adrian Osorio, Caleb Smith, Anthony Drabeck, Fadi Samaan, Jun. 2020.
- “Mobile Simulated Buoy Hub II,” Shyam Patel, Joseph Parra, Haniel Youlesivanson, Jeffery Vuong, and Alexander Kim, Mar. 2020.
- “Mobile Simulated Buoy Hub I,” Taylor Filemon, Chunguang Yang, Jiayi Wu, and Christopher Navarro, Mar. 2020.
- “Unmanned Surface Vehicle with Auto Attach/Detach Data Connectors,” Lorjean Sagabaen, Alexander Janbo, Dennis Chua, Yuchen Geng, and James Le, Mar. 2020.
- “Autonomous Target Scoring Drone II,” Leo Leal, Kian Farsany, June Chen, and Zachary Agness, Mar. 2020.
- “Trilateral FM Signal Locator,” Yiran Xu, Gary Zhou, and Zhi Qiu, Mar. 2020.
- “Wideband Spectrum Monitor,” Brian Dang, Mark Tullen, Jesus Lopez, and Vincent Villacorta, Mar. 2020.
- “Rogue RF Signal Detector Drone,” Yehuda Rousso, Erick Magana, and Che-Wei Chang, Mar. 2020.
- “Drone Leader-Follower II,” Andy Vu, Lawrence Dizon, David Phan, and Hugh Fong, Mar. 2020.
- “Autonomous Target Robot II,” Olga Shigapova, Don Isbell, and Carmine Choi, Mar. 2020.
- “Drone Leader-Follower I,” Ameer Hussain, Amauri Villegas, Phillip Flores, and

Jacob Hurley, Mar. 2020.

- “Autonomous Target Robot I,” Alex Yamamura, Brandon Kuo, Simon Lee, and William Hartono, Mar. 2020.
- “Autonomous Target Scoring Drone I,” Sukhmanjit Kaur, Eric Rodriguez, Bert Yu, and Bibek Adhikari, Mar. 2020.
- “Auto-Adjusting P2P Microwave Link,” David Tiao, Wilfredo Paraiso, and Wilfredo Paraiso, Mar. 2020.
- “Rogue RF Signal Detector Drone,” Sem Diaz, Edwin Torres, and Matthew Scudder, Mar. 2019.
- “Trilateral FM Signal Locator,” Sean Pickman and Alan Nguyen, Mar. 2019.
- “Wideband Spectrum Monitor,” Daniel Aleman, Robert Almeida, and Karla Penney, Mar. 2019.
- “Autonomous Target Robot,” Coulter Mulvihill, Henry Hua, and Xinyi Wu, Mar. 2019.
- “Auto-Adjusting P2P Microwave Link,” Anthony Valverde, Alex Roissier, and Robert Wells, Mar. 2019.
- “Autonomous Target Scoring Drone,” Morgan Lytle, Kevin Le, and Alec Pflaume, Mar. 2019.
- “Autonomous Aerial Position Tracking System,” Gurkaran Randhawa, Jack Maynard, and Estevan Valencia, Mar. 2019.
- “Autonomous Target Robot,” Taylor Che, Samuel Choi, and Francisco Munoz, Mar. 2018.
- “Rogue Radio Frequency (RF) Signal Detector,” Francisco Raygoza, Isai Rocha, and Victor Medel, Mar. 2018.
- “Auto-Adjusting P2P Microwave Link,” Shane Owens, Jesse Layman, and Curtis Smith, Mar. 2018.
- “Autonomous Target Scoring Drone,” Jiaming Zhou and Gerardo Hernandez, Mar. 2018.
- “Trilateral FM Signal,” Wang Hu, Rongjia Zhang, and Xin Ke, Mar. 2018.
- “Localization of RF Transmission Using Phased Arrays,” Albert Liu and Matthew Lumantas, Mar. 2018.
- “Wideband Spectrum Monitor,” Christian Ardito and Bradley Evans, Mar. 2018.

**Ph.D.
Dissertation
Defense
Committees**

- Russell Whiton, “Dude, Where’s my Car? - Cellular Navigation for Autonomous Vehicles,” Lund University, Sweden, Apr. 2024
- Joshua Critchley-Marrows, “A Return to the Stars in an Age of Navigation Uncertainty – A Treatise on the Use of Celestial Bodies for Resilient Positioning, Navigation and Timing,” University of Sydney, Australia, Feb. 2024
- Yiming Chen, “A New Approach for Computationally Efficient and Reliable Carrier Integer Ambiguity Resolution in GPS/INS,” University of California, Riverside, USA, Dec. 2014.
- Sheng Zhao, “Realtime, Decimeter Accuracy Navigation Using Sliding Window Estimator and Autonomous Vehicle Trajectory Tracking Control,” University of California, Riverside, USA, Dec. 2014.

**M.S. Thesis
Committees**

- Jiarui Liu, “Analysis of EV Charging Load Based on Household Driving Data in California,” Jul. 2015.

Outreach

Creator of “Game of Drones,” an annual competition for middle and high school students to inspire students to pursue science, technology, engineering, and mathematics (STEM) fields. Since 2016, Game of Drones has engaged 300+ students.